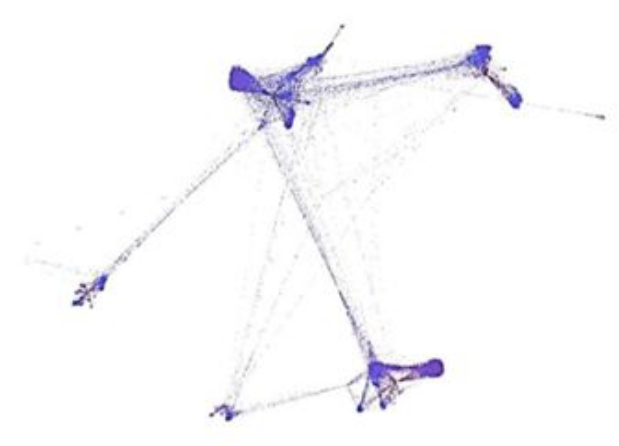
Using Graph Theory to Model Disease Spread in Populations

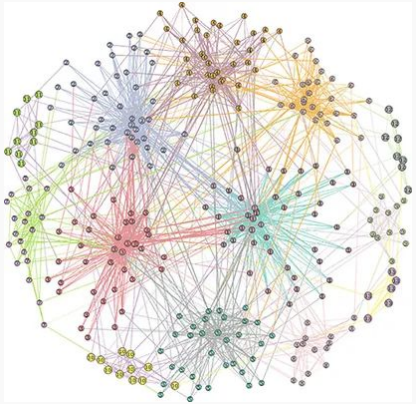
Grace Wilding

Graphs

Graphs are mathematical structures used to visualize relationships between entities in a network. Graph Theory is the field that studies this structures.



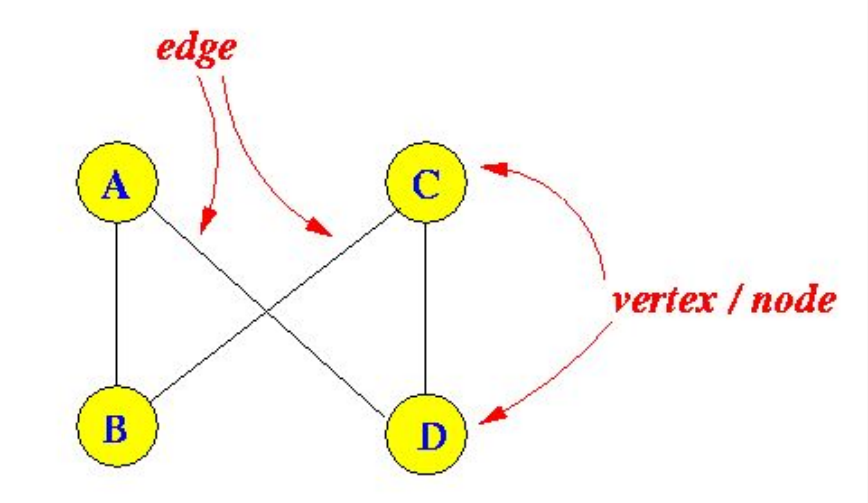
Kumar and Sinha (2021); see references.

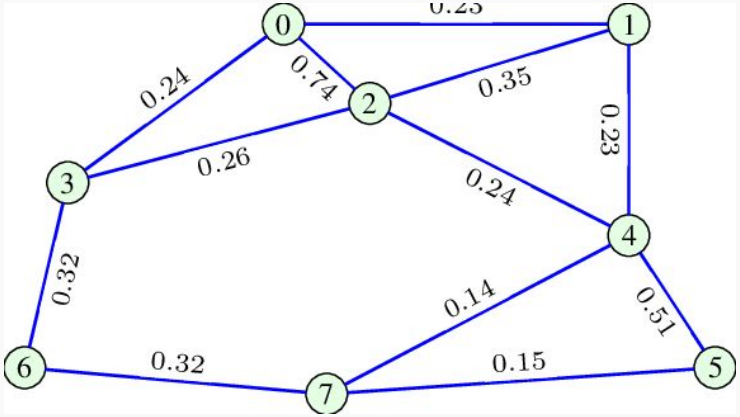


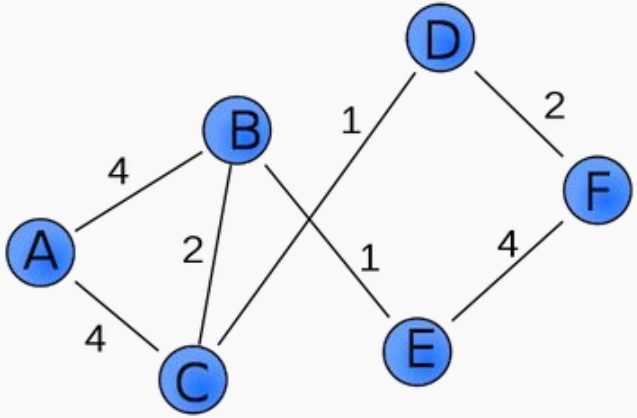
Nicolini, C., Bifone, A., & Bordier, C. (2017, August 3). *Graph Analysis and Modularity of Brain Functional Connectivity Networks: Searching for the Optimal Threshold.* Retrieved December 5, 2025, from https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2017.00441/full

Edges, Vertices, and Degree

- A **vertices/nodes** represent people, items, groups, or any objects.
- An **edge** connects two vertices together showing a relationship between the vertices.
    - In a disease network, edges represent opportunities for disease transmission.
- The **degree of a vertex** is the number of edges a vertex has
    - Example: Vertex A has a degree of 2.







Edge weighting

Edges can be assigned **weights** which are numbers representing the strength or nature of a relationship.

Depending on the context, weights can represent (but not limited to):

- Frequency of contact
- Intensity of contact/relationship
- Probability of transmission
- Distance

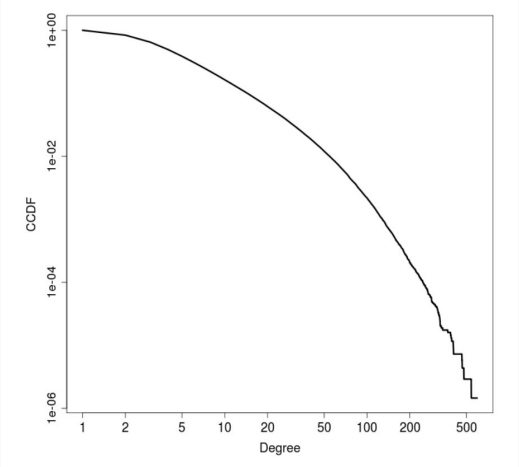
Weighted graphs allow for the network to contain for information for deeper analysis.

Degree Distributions

Degree distributions is the probability distribution of different degrees in the network. Each probability is found using the following:

$$P(K) = \frac{N_K}{N}$$

Where N_K is the number of vertices with degree K and N is the total number of vertices.



Degree Distribution [Photograph]. (n.d.). https://www.sci.unich.it

The graph I will build

I will demonstrate how Enns and Brandeau (2011) built a network and inferred the model parameters in their article, “Inferred model parameters in network-based disease simulation” representing a HIV disease network in sub-Saharan Africa.

- Vertices -> individuals in the population
- Edges -> sexual relationships
- Partnership types: spousal and non-spousal
- Degree distributions inferred from demographic surveys
- Should model real population’s behavior

Note: Transmission will only be through heterosexual contact in this network to maintain simplicity (HIV can also be spread through needle sharing and homosexual contact but accounting for this would make the network too complex).

Assigning Edges

Using the survey information to draw the edges:

1. **Spousal relationships** are drawn first, using census data.
2. **Non-spousal relationships** range from 0 to 4 per node. These are drawn using a degree distribution to match the percentage of those who reported 0, 1, 2, 3, or 4 partners in a survey the researchers conducted.

Why is this a good choice?

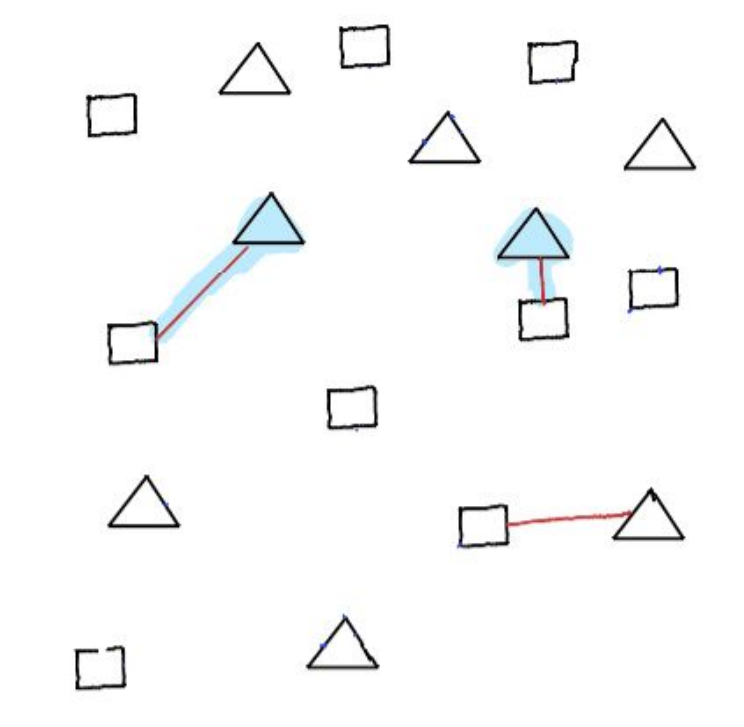
Example of Assigning Edges

In this network, there will be 8 men and 9 women. Using the survey results, the degree distributions for each gender is below:

# of Partners / Degree	Men	Women
1	25.0% (2)	33.3% (3)
2	50.0% (4)	44.4% (4)
3	12.5% (1)	22.2% (2)
4	12.5% (1)	0.0% (0)

Census data: 3 marriages in this population

Example cont.

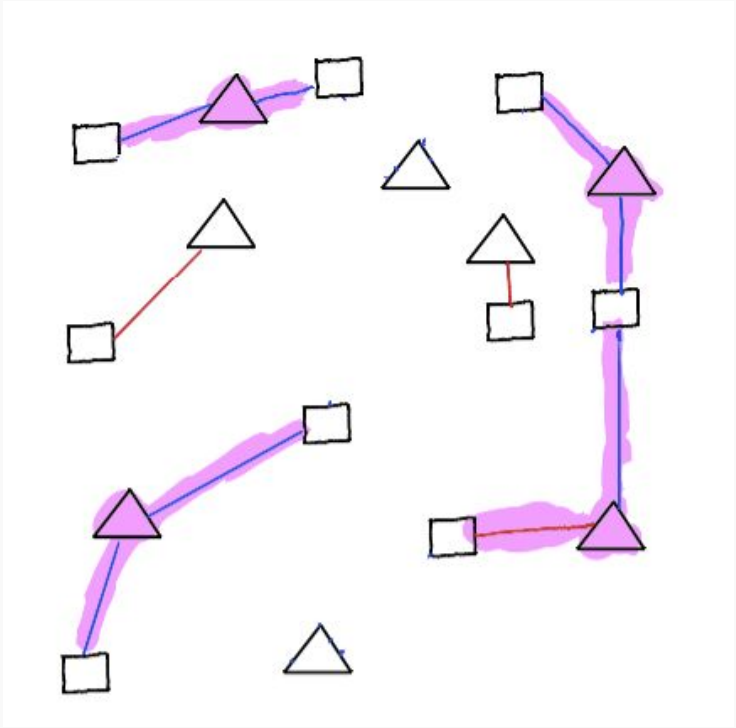


Using the census data three pairs are found at random to represent the three marriages.

Only 25.0% of men reported only one partner, so 2 out of 3 of the men in marriages will maintain one partner.



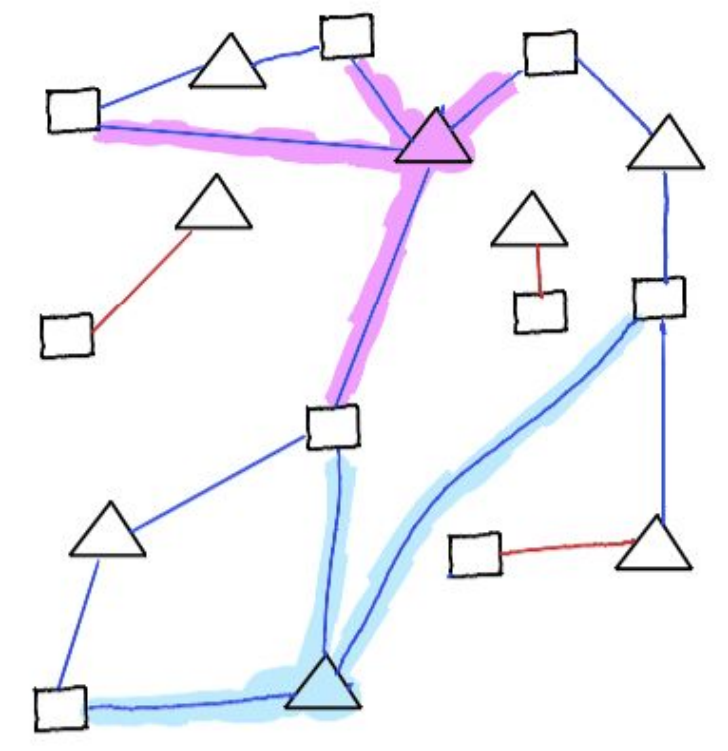
Example cont.



The men in the network are picked at random, excluding those designated to have one partner, to match the 50.0% of men who have two partners.



Example cont.



Then the edges one male who reported having three partners are drawn out. Finally the edges for the last male who had four relationships are drawn out.

The degrees of all of the female vertices are checked if they match the female degree distribution. Adjustments to the graph can be made if needed.



Finding the Probability of Transmission

In order to find how the network changes over time, the probability of transmission of HIV needed to be calculated for pairs of vertices. The probability was based on:

- **Disease Stage**: acute, chronic, in treatment
- **Partnership Type**: spousal, non-spousal

The model is simulated over time using these probabilities and the probability of relationship dissolution using known data.

Analysis

Using edge weights:

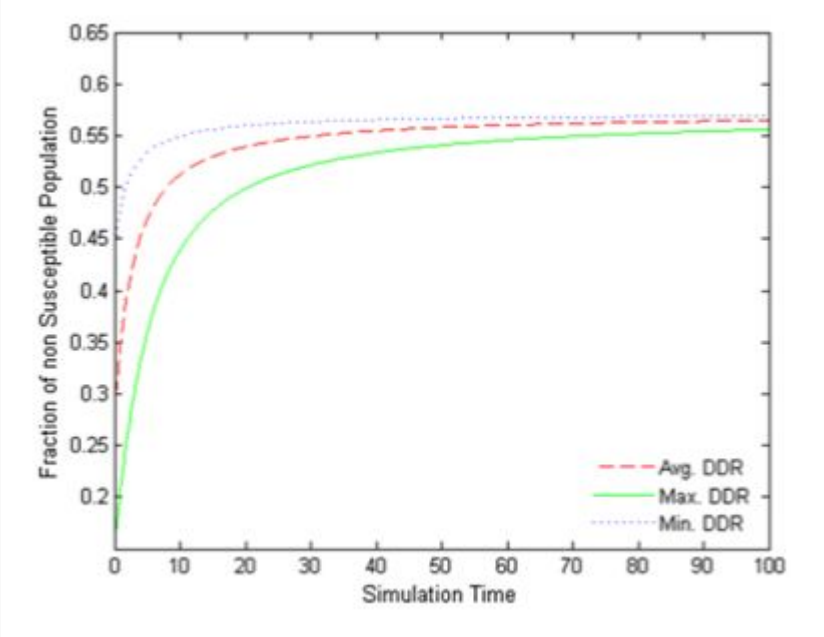
- The spread of HIV can be simulated over time using MatLab
- Researchers can compare baseline spread to studies where:
  - HIV prevention medication is used
  - Each subject goes through a HIV educational program
  - Other models

If more detailed information is collected, the graph could also help with contact tracing to help identify individuals who have been exposed to HIV.

Simulating Spread Over Time

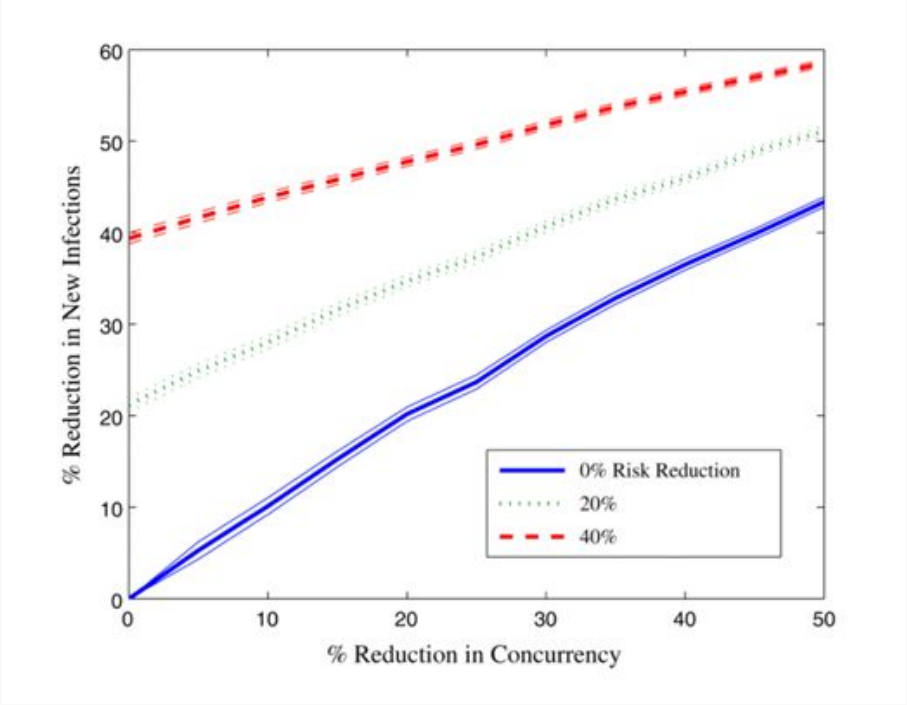
This was a graph used in analyzing a social network, but this is something that can be done in the HIV network.

For the HIV network it would the fraction of people free of HIV as time passes.



Kumar and Sinha (2021); see references.

Comparative Analysis



The graph shows the impact of two HIV-prevention strategies in Tanzania: reducing concurrent partnerships (discouraging having more than one partner at a time) and promoting safer sex practices such as consistent condom use. It compares different levels of “risk prevention,” which represent how much these safer-sex campaigns reduce the probability of HIV transmission in non-spousal partnerships.

Conclusions

Graph Theory provides powerful tools for modeling and understanding disease spread.

When building a network, researchers often have to decide on two components to construct the graph:

1. Identifying edges
2. Using probabilities to determine the change over time

Graphs allow researchers to analyze disease spread, evaluate prevention strategies, and improve public health interventions. This paper focused on a HIV network, but this can be applied to other diseases using similar strategies.

References

Enns, E. A., & Brandeau, M. L. (2011, June). *Inferring model parameters in network-based disease simulation*. Retrieved December 5, 2025, from https://pubmed.ncbi.nlm.nih.gov/21373984/

Kumar, P., & Sinha, A. (2021, January 9). *Information diffusion modeling and analysis for socially interacting networks*. Retrieved December 5, 2025, from https://pubmed.ncbi.nlm.nih.gov/33456625/

Thank you for listening!

Any Questions?